

Laboratorio de Microeconomía I
Centro de Investigación y Docencia Económicas
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Tarea 7

Exercises

Ex. 1.38 Verify that the expenditure function obtained from the CES direct utility function in **Example 1.3** (J&R, 2011; p.39) satisfies all the properties given in **Theorem 1.7**.

Ex. 1.56 What restrictions must the α_i , $f(y)$, $w(p_1, p_2)$, and $z(p_1, p_2)$ satisfy if each of the following is to be a legitimate indirect utility function?

- (a) $v(p_1, p_2, p_3, y) = f(y)p_1^{\alpha_1}p_2^{\alpha_2}p_3^{\alpha_3}$
 - (b) $v(p_1, p_2, y) = w(p_1, p_2) + z(p_1, p_2)/y$
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Ex. 1.59 If $e(p, u) = z(p_1, p_2)p_3^m u$, where $m > 0$, what restrictions must $z(p_1, p_2)$ satisfy for this to be a legitimate expenditure function?

Ex. 3.E.8^A For the Cobb–Douglas utility function, verify that the relationships in (3.E.1) and (3.E.4) hold. Note that the expenditure function can be derived by simply inverting the indirect utility function, and vice versa.

Ex. 3.G.15^B Consider the utility function

$$u = 2x_1^{1/2} + 4x_2^{1/2}.$$

- (a) Find the compensated demand function $h(\cdot)$ (Hicksian demand functions).
 - (b) Find the expenditure function, and verify that $h(p, u) = \nabla_p e(p, u)$.
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Ex. (Opcional) For the utility function in **Exercise 3.G.15^B**, solve the following:

- (c) Using **Theorem 1.8**, derive the indirect utility function using the expenditure function from (b).
 - (d) Using **Theorem 1.9**, derive the Marshallian demand function for good x_1 using the Hicksian demand function you derived in (a) and the indirect utility function you derived in (c).
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Appendix

We define the **expenditure function** as the minimum-value function,

$$e(\mathbf{p}, u) \equiv \min_{\mathbf{x} \in \mathbb{R}_+^n} \{\mathbf{p} \cdot \mathbf{x} \mid u(\mathbf{x}) \geq u\} \quad (1.14)$$

for all $\mathbf{p} \gg \mathbf{0}$ and all attainable utility levels u . For future reference, let $\mathcal{U} = \{u(\mathbf{x}) \mid \mathbf{x} \in \mathbb{R}_+^n\}$ denote the set of attainable utility levels. Thus, the domain of $e(\cdot)$ is $\mathbb{R}_{++}^n \times \mathcal{U}$.

Note that $e(\mathbf{p}, u)$ is well-defined because for $\mathbf{p} \in \mathbb{R}_{++}^n$, $\mathbf{x} \in \mathbb{R}_+^n$, $\mathbf{p} \cdot \mathbf{x} \geq 0$. Hence, the set of numbers $\{e \mid e = \mathbf{p} \cdot \mathbf{x} \text{ for some } \mathbf{x} \text{ with } u(\mathbf{x}) \geq u\}$ is bounded below by zero. Moreover because $\mathbf{p} \gg \mathbf{0}$, this set can be shown to be closed. Hence, it contains a smallest number. The value $e(\mathbf{p}, u)$ is precisely this smallest number. Note that any solution vector for this minimisation problem will be non-negative and will depend on the parameters $\mathbf{p}$ and u . Notice also that if $u(\mathbf{x})$ is continuous and strictly quasiconcave, the solution will be unique, so we can denote the solution as the function $\mathbf{x}^h(\mathbf{p}, u) \geq \mathbf{0}$. As we have seen, if $\mathbf{x}^h(\mathbf{p}, u)$ solves this problem, the lowest expenditure necessary to achieve utility u at prices $\mathbf{p}$ will be exactly equal to the cost of the bundle $\mathbf{x}^h(\mathbf{p}, u)$, or

$$e(\mathbf{p}, u) = \mathbf{p} \cdot \mathbf{x}^h(\mathbf{p}, u). \quad (1.15)$$

Theorem 1.7 Properties of the Expenditure Function.

If $u(\cdot)$ is continuous and strictly increasing, then $e(\mathbf{p}, u)$ defined in (1.14) is

1. Zero when u takes on the lowest level of utility in $\mathcal{U}$,
2. Continuous on its domain $\mathbb{R}_{++}^n \times \mathcal{U}$,
3. For all $\mathbf{p} \gg \mathbf{0}$, strictly increasing and unbounded above in u ,
4. Increasing in $\mathbf{p}$,
5. Homogeneous of degree 1 in $\mathbf{p}$,
6. Concave in $\mathbf{p}$.

If, in addition, $u(\cdot)$ is strictly quasiconcave, we have

7. *Shephard's lemma*: $e(\mathbf{p}, u)$ is differentiable in $\mathbf{p}$ at $(\mathbf{p}^0, u^0)$ with $\mathbf{p}^0 \gg \mathbf{0}$, and

$$\frac{\partial e(\mathbf{p}^0, u^0)}{\partial p_i} = x_i^h(\mathbf{p}^0, u^0), \quad i = 1, \dots, n.$$

Theorem 1.8 Relations Between Indirect Utility and Expenditure Functions.

Let $v(\mathbf{p}, y)$ and $e(\mathbf{p}, u)$ be the indirect utility function and expenditure function for some consumer whose utility function is continuous and strictly increasing. Then for all $\mathbf{p} \gg \mathbf{0}$, $y \geq 0$, and $u \in \mathcal{U}$:

1. $e(\mathbf{p}, v(\mathbf{p}, y)) = y$.
2. $v(\mathbf{p}, e(\mathbf{p}, u)) = u$.

Theorem 1.9 Duality Between Marshallian and Hicksian Demand Functions.

Under Assumption 1.2 we have the following relations between the Hicksian and Marshallian demand functions for $\mathbf{p} \gg \mathbf{0}$, $y \geq 0$, $u \in \mathcal{U}$, and $i = 1, \dots, n$:

1. $x_i(\mathbf{p}, y) = x_i^h(\mathbf{p}, v(\mathbf{p}, y))$.
2. $x_i^h(\mathbf{p}, u) = x_i(\mathbf{p}, e(\mathbf{p}, u))$.

Proposition 3.E.1

Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated preference relation $\succsim$ defined on the consumption set $X = \mathbb{R}_+^L$ and that the price vector is $p \gg 0$. We have

- (i) If x^* is optimal in the UMP when wealth is $w > 0$, then x^* is optimal in the EMP when the required utility level is $u(x^*)$. Moreover, the minimized expenditure level in this EMP is exactly w .
- (ii) If x^* is optimal in the EMP when the required utility level is $u > u(0)$, then x^* is optimal in the UMP when wealth is $p \cdot x^*$. Moreover, the maximized utility level in this UMP is exactly u .

Proposition 3.E.1 allows us to make an important connection between the expenditure function and the indirect utility function developed in Section 3.D. In particular, for any $p \gg 0$, $w > 0$, and $u > u(0)$ we have

$$e(p, v(p, w)) = w \quad \text{and} \quad v(p, e(p, u)) = u. \quad (3.E.1)$$

These conditions imply that for a fixed price vector $\bar{p}$, $e(\bar{p}, \cdot)$ and $v(\bar{p}, \cdot)$ are inverses to one another (see Exercise 3.E.8) (...) there is a direct correspondence between the properties of the expenditure function and the indirect utility function. They both capture the same underlying features of the consumer's choice problem.

Using Proposition 3.E.1, we can relate the Hicksian and Walrasian demand correspondences as follows:

$$h(p, u) = x(p, e(p, u)) \quad \text{and} \quad x(p, w) = h(p, v(p, w)). \quad (3.E.4)$$